

4

Case Studies on the Initiatives and Success Stories of Edge AI Systems for Agriculture

Thamizhiniyan Natarajan and Shanmugavadivu Pichai
Gandhigram Rural Institute, Chinnalapatti, India

4.1 INTRODUCTION

An era of unprecedented transformation is being ushered into the agricultural sector, driven by the fusion of cutting-edge technology with age-old farming practices, spearheaded by the advent of Edge AI (Artificial Intelligence at the Edge). Within this dynamic landscape, Edge AI is positioned as a catalyst for revolutionizing agriculture, offering a multitude of solutions to address long-standing challenges and usher in a new era of smart farming.

For millennia, agriculture has served as the bedrock of human survival, providing sustenance, livelihoods, and economic prosperity to nations across the globe. However, the demands on this critical sector have never been more significant. With the world's population surging and climate change disrupting traditional weather patterns, agriculture faces mounting pressure to meet the growing need for food production while simultaneously contending with limited resources and environmental degradation.

The role of Edge AI in agriculture becomes apparent as the convergence of artificial intelligence and the Internet of Things (IoT) empowers farms with real-time insights, facilitating immediate responses to changing conditions and enabling data-driven precision agriculture.

The pivotal role of Edge AI in agriculture is highlighted through a comprehensive exploration of a curated collection of case studies drawn from diverse research papers. These case studies encapsulate the transformative potential of Edge AI, showcasing its ability to reshape the farming landscape in multifaceted ways.

From the utilization of AI-powered cameras for detecting plant diseases in the field to the remote monitoring of livestock health through edge devices, from precision irrigation systems that fine-tune water distribution based on real-time data to autonomous farming solutions that minimize manual labor while maximizing productivity, a spectrum of agricultural applications is covered. The functional features and benefits of Edge AI systems are illuminated, and the socio-economic impact, adoption, and implications for farming communities and the environment are delved into.

As this journey through the realm of Edge AI in agriculture is embarked upon, it becomes evident that more than a technological evolution is being witnessed. A fundamental shift is underway, one that promises to redefine the very nature of farming itself. Empowering farmers with tools for informed decision-making, optimizing resource allocation, minimizing environmental footprints, and holding the key to global food security in a world marked by uncertainties are the promises of Edge AI.

In each case study, stories of innovation, resilience, and transformation are unveiled. These narratives underscore the profound influence of Edge AI on agriculture, reinforcing its status as a pivotal force in shaping the future of food production, promoting sustainability, fostering innovation, and ensuring food security in an increasingly complex and interconnected world. As the details of each case study are delved into, an invitation is extended to witness first-hand how Edge AI is turning the age-old practice of farming into a cutting-edge science.

The chapter's structure unfolds as follows: Section 4.2 explores related works on Edge AI in agriculture, offering insights into the field's existing research. In Section 4.3, an extensive examination of case studies is presented, offering a deeper understanding of practical applications. Section 4.4 comprehensively addresses the utilization, benefits, socio-economic impact, and adoption of Edge AI in agriculture. Lastly, Section 4.5 serves as the conclusion, summarizing the chapter's key findings and insights.

4.2 RELATED WORK

In a world driven by data and constant innovation, the combination of artificial intelligence (AI) with the untapped potential of edge computing and the IoT is recognized as a revolutionary force ready to transform industries, societies, and our daily lives. AI, known for its ability to uncover patterns, predict outcomes, and automate tasks, has become an integral part of modern existence. From recommendation algorithms that guide our entertainment choices to autonomous vehicles navigating busy streets, AI silently orchestrates the marvels of our time.

However, as AI's influence proliferated, an inherent challenge materialized. Traditional AI systems, predominantly housed in remote data centers and cloud infrastructures, dazzled with their prowess but were hampered by latency issues. The delay between data generation and its subsequent analysis, while insignificant for specific use cases, emerged as a critical factor in situations requiring immediate decision-making. This is where the concept of “edge” becomes relevant [1, 2].

4.2.1 The Edge AI Advantage

Edge AI signifies a radical departure from the centralized AI paradigm. It positions computation, data storage, and processing at the very periphery of the network – right where the action unfolds. By doing so, it collapses the time lag between data generation and analysis, ushering in a new era of immediacy and responsiveness [1]. The advantages of Edge AI are as follows:

- *Real-Time Decision-Making*: One of the foremost advantages of Edge AI is its capacity for real-time decision-making. Devices equipped with Edge AI can process data instantaneously, enabling split-second responses. This capability is a game-changer for applications like autonomous vehicles, industrial automation, and health-care monitoring, where even the briefest delay can have significant consequences [3].
- *Enhanced Privacy and Security*: Edge AI also bolsters data privacy and security. Since data is processed locally on the device, sensitive information need not traverse networks, reducing the risk of exposure. This is particularly vital in applications like health care, where patient data must be handled with utmost confidentiality [4].

- *Bandwidth Efficiency*: Edge AI optimizes bandwidth usage by processing data locally. This reduces the strain on network infrastructure, making it suitable for IoT deployments in remote or bandwidth-constrained environments [5].
- *Scalability*: Edge AI exhibits remarkable scalability, making it suitable for an extensive array of devices and usage scenarios. It can be customized to align with the distinct requirements of various applications, spanning from small sensors to robust industrial machinery [6].

4.2.2 The Evolution of Edge AI

The concept of Edge AI didn't emerge overnight but evolved over the years as a response to the growing demands of real-time, distributed computing.

Historically, AI relied on centralized computing resources, with data sent to remote servers for processing. However, as AI applications expanded into areas like autonomous vehicles, industrial automation, and IoT, the limitations of this approach became evident. The latency introduced by round-trip data transfer to distant data centers was unacceptable in scenarios where immediate decisions were crucial.

The concept of "edge computing" emerged to define a novel approach where computing resources were positioned nearer to the data source within the network. This transformation laid the foundation for the inception of Edge AI [1]. Leveraging advancements in hardware capabilities, Edge AI devices gained the capacity to execute intricate AI computations locally.

In agriculture, for instance, Edge AI-enabled tractors and drones to analyze field data in real-time, optimizing planting and harvesting decisions. In health care, wearable devices equipped with Edge AI became capable of monitoring vital signs and alerting users to potential health issues without requiring constant cloud connectivity.

4.2.3 Edge AI and IoT in Agriculture

Edge AI's impact extends into the agricultural sector, ushering in a new era of precision farming and sustainable practices.

- *Precision Agriculture*: Edge AI empowers agricultural machinery with the capability of making immediate decisions regarding planting,

harvesting, and irrigation. Through sensors and cameras installed on tractors and drones, data related to soil conditions, crop health, and weather patterns is gathered. This information is promptly processed by Edge AI, allowing farmers to enhance their operations, minimize resource utilization, and boost crop yields [7].

- *Crop Monitoring*: Edge AI-powered cameras placed throughout fields can monitor crop health, identifying signs of disease, pest infestations, or nutrient deficiencies. This data can be used to trigger automated responses, such as targeted pesticide application, thereby reducing the need for broad-spectrum treatments and minimizing environmental impact [8].
- *Livestock Management*: In livestock farming, Edge AI can monitor the health and behavior of animals in real time. Wearable sensors and cameras can detect signs of distress or illness, allowing for early intervention. Additionally, AI-powered feeders can dispense precise amounts of feed, reducing waste and optimizing nutrition [9].

4.2.4 Challenges and Considerations

While Edge AI offers tremendous potential in agriculture, it comes with its own set of challenges:

- *Connectivity*: In remote agricultural areas, connectivity can be limited. Reliable network infrastructure is essential to ensure seamless data transfer between edge devices and central systems [1].
- *Data Privacy*: Edge AI collects a vast amount of data, including information about crop conditions and farm operations. Ensuring the privacy and security of this data is paramount [10].
- *Scalability*: The agricultural sector encompasses a wide range of operations, from small family farms to large commercial enterprises. Edge AI solutions must be scalable and adaptable to meet the needs of diverse agricultural settings [10].

In the world of Edge AI, there are endless opportunities for innovation. It's a space where intelligence operates right at the edge of what's possible, where data quickly turns into useful information, and where Edge AI's history and future are closely intertwined.

4.3 TRANSFORMATIVE TALES: CASE STUDIES SHOWCASING EDGE AI IN AGRICULTURE

In this section, the focus shifts toward real-world applications of Edge AI within the agricultural sector. A series of case studies is delved into, demonstrating how Edge AI systems have been effectively put into operation to tackle pressing challenges and promote innovation in farming practices. Both national and international contexts are covered by these case studies, underlining the versatility and transformative potential of Edge AI in agriculture. In each instance, attention is drawn to the unique approaches taken to leverage Edge AI technologies for the optimization of operations, enhancement of crop yields, improved resource management, and, ultimately, the contribution to a more sustainable and productive agricultural landscape. These case studies stand as concrete examples of the capability of Edge AI to revolutionize farming, thereby paving the way for a future in agriculture that is smarter, more environmentally friendly, and highly efficient.

4.3.1 Enhancing Plant Disease Classification with Edge AI on IoT Devices

In the pursuit of more efficient and sustainable farming practices, Trong-Minh Hoang and their team [11] have introduced a ground-breaking approach to plant disease classification using deep convolutional neural networks (DCNN) on IoT edge devices. Traditional methods for disease detection in agriculture have often been limited in their scope and efficiency. To address these limitations, the researchers have developed a lightweight and highly efficient DCNN model that is well-suited for deployment on resource-constrained IoT devices. This case study underscores the pivotal role of edge computing and AI solutions in the realm of precision agriculture, shedding light on how machine-learning algorithms can significantly enhance the timely and accurate detection of diseases in smart farming.

4.3.1.1 Challenges Addressed

- *Resource Constraints:* Edge devices, typically employed in remote farming areas, often have limited computational resources. This constraint hampers the utilization of complex machine-learning models, which are essential for accurate disease identification.

- *Model Size*: Downsizing machine-learning models to fit edge devices without sacrificing accuracy is a complex problem. Shrinking models too aggressively can lead to a loss of critical information and a decrease in identification accuracy.
- *Real-World Applicability*: Models must perform well not only in laboratory conditions but also in real-world agricultural settings, where environmental factors can introduce significant variability.

4.3.1.2 Key Benefits

- *Real-Time Insights*: Edge AI enables farmers to receive real-time updates on the health of their crops, allowing for immediate intervention when disease outbreaks occur.
- *Proactive Disease Management*: With timely detection, farmers can take proactive measures to prevent the spread of diseases, minimizing crop losses.
- *Improved Yields*: By addressing diseases promptly, farmers can optimize their agricultural practices and enhance overall crop yields.
- *Resource Efficiency*: The efficiency of the DCNN model ensures that it can operate effectively on resource-constrained IoT devices, making it a practical solution for real-world applications in precision agriculture.

4.3.1.3 Results and Implications

The DCNN model developed by Trong-Minh Hoang and their team has demonstrated remarkable accuracy in plant disease classification, outperforming other machine-learning models while maintaining high efficiency. This case study serves as a testament to the potential of machine-learning algorithms in achieving timely and accurate disease detection on IoT edge devices. Furthermore, it highlights the transformative impact of Edge AI in revolutionizing the agriculture industry and bolstering global food security.

4.3.2 Securing Smart Climate Agriculture with Edge AI and Blockchain Integration

In the modern era, technology plays a pivotal role in advancing various sectors, including agriculture. To address the unique challenges faced by agriculture, especially in countries like India, there is a growing need for technological modernization. This case study, inspired by the work of Li

Ting et al. [12], explores the development of an intelligent agricultural system that leverages cutting-edge technology to monitor and optimize essential climate and environmental parameters. By integrating IoT and blockchain technology, this system offers efficient and secure decision-making capabilities, ultimately enhancing plant health and agricultural productivity.

4.3.2.1 Challenges Addressed

- *Data Privacy and Security:* With the increasing number of IoT devices connecting to agricultural systems, the case study prioritizes data privacy and security. It accomplishes this by implementing blockchain technology to guarantee the secure transmission and storage of sensitive data.
- *Real-Time Monitoring:* Successful agriculture requires continuous monitoring of crucial parameters, including temperature, humidity, soil moisture, and light intensity. The proposed system integrates IoT sensors to collect and transmit this data in real time, enabling prompt analysis and decision-making.
- *Water Management:* Water conservation is essential in agriculture. The system employs intelligent fuzzy logic to make well-informed decisions regarding the timing and quantity of irrigation, ensuring efficient water usage and optimal plant health.

4.3.2.2 Key Benefits

- *Real-Time Decision-Making:* The integrated system empowers farmers with real-time insights into crucial environmental parameters. By analyzing this data and applying intelligent fuzzy logic, the system can make informed decisions to meet the watering requirements of plants promptly.
- *Enhanced Security:* Blockchain technology is utilized to enhance the security of the IoT-enabled system. This ensures that only trusted devices have access to the network, mitigating potential security threats.
- *Remote Monitoring:* Multiple users can remotely monitor and interact with the system simultaneously using an Android application. This remote accessibility allows for efficient management of agricultural operations.
- *Water Efficiency:* The system not only notifies users about watering requirements but also provides alerts for turning water motors on and off. This feature contributes to efficient water resource management.

4.3.2.3 Results and Implications

The experimental outcomes of the proposed system, as demonstrated by Li Ting et al.'s research, demonstrate its scalability, efficiency, and security. It effectively monitors and optimizes climate and environmental parameters, contributing to plant health and increased agricultural productivity. While the current system focuses on eight types of plants, future directions could involve implementing advanced techniques like genetic algorithms or neural networks to provide even more precise and accurate recommendations. This case study highlights the transformative potential of Edge AI and blockchain integration in agriculture, showcasing how modern technologies can enhance efficiency, productivity, and sustainability in this vital sector.

4.3.3 Enhancing Data Collection Efficiency in Smart Agriculture through Edge Computing

The utilization of wireless sensor networks (WSNs) in agriculture has brought about noteworthy progress in data-centric farming techniques. Nevertheless, conventional agricultural WSNs frequently grapple with problems such as duplicated data and prolonged response times when confronted with critical events (CEs). These hurdles lead to amplified time and energy usage, impeding the possible advantages of precision agriculture. To tackle these predicaments, an inventive data collection technique, driven by edge computing (EC), has been introduced by Xiaomin Li et al. [13] to address CEs in the realm of smart agriculture. This case study delves into the methodology and results of this innovative approach.

4.3.3.1 Challenges Addressed

- *Data Redundancy*: Traditional WSNs generate large volumes of redundant data, leading to inefficiencies in data collection, storage, and analysis.
- *High Latency*: CEs in agriculture require real-time data collection and decision-making. High latency in data transmission can hinder timely responses, impacting crop health and yield.

4.3.3.2 Key Benefits

The proposed approach offers several key benefits:

- *Data Reduction*: By extracting and prioritizing key features data types (KFDTs) from historical datasets, the approach minimizes redundant data, optimizing data collection efficiency.
- *Low Latency*: Through EC, event types are swiftly identified based on minimum average variance, enabling rapid data collection by sensing nodes within latency constraints.
- *Resource Efficiency*: By reducing the number of required sensors, minimizing sensing time, optimizing data collection volume, and streamlining communication time, the approach enhances resource utilization efficiency.

4.3.3.3 Results and Implications

A real-world testbed within a smart greenhouse was established to validate the approach proposed by Xiaomin Li et al. The results underscored its effectiveness in reducing data redundancy, achieving low-latency data collection, and optimizing resource utilization. In comparison to conventional methods, the proposed strategy notably enhanced the equilibrium between data validity, energy consumption, and latency.

4.3.4 Enhancing Precision Agriculture with Low-Power Edge AI

In the domain of precision agriculture, characterized by the utmost importance of efficient resource utilization and crop optimization, the incorporation of AI has surfaced as a highly promising pathway to enhance farming methods. This case study explores pioneering work presented by Dmitrii Shadrin et al. [14] focusing on the detection of seed germination using AI within a low-power embedded system, offering insights into the transformative potential of Edge AI in agriculture.

4.3.4.1 Challenges Addressed

- *Low-Power AI Integration*: The integration of AI into resource-constrained embedded systems is a challenge, and this study aims to demonstrate the feasibility and benefits of such integration in agricultural contexts.
- *Efficient Seed Germination Detection*: Traditional seed germination monitoring methods may lack efficiency and accuracy. The paper addresses the need for a more effective approach using AI and computer vision.

4.3.4.2 *Key Benefits*

- *Resource-Efficient AI:* The study showcases the development of a low-power sensing system equipped with AI capabilities. This innovation allows for AI processing at the edge, reducing the reliance on cloud-based solutions and minimizing data transmission requirements.
- *Enhanced Germination Detection:* By implementing a Convolutional Neural Network (CNN), the system presented in this study achieves impressive results, boasting an average Intersection over Union (IoU) score of 83% on the test dataset and 97% accuracy in seed recognition on the validation dataset. Such a high level of accuracy contributes significantly to the detection of seed germination.

4.3.4.3 *Results and Implications*

In practice, the low-power embedded system equipped with AI capabilities can independently detect seed germination dynamics without relying on extensive data transmission to cloud servers. This not only cuts down on data-related expenses but also guarantees real-time monitoring and decision-making.

Furthermore, the scalability of this approach opens doors to widespread applications in precision agriculture. It enables the assessment of growing systems' performance, predictions about harvesting periods, and opportunities for resource optimization during the early stages of plant growth. This optimization, driven by AI insights, contributes to more efficient resource management in precision agriculture.

The study's experimental comparison between the low-power embedded system and a desktop computer highlights the advantages of Edge AI. While the desktop computer may be significantly faster, the low-power embedded system's ability to operate on battery power makes it suitable for remote and autonomous IoT applications, aligning perfectly with the emerging trends in smart agriculture.

4.3.5 **Revolutionizing Precision Agriculture with Edge AI Sensor Virtualization**

In the ever-changing realm of precision agriculture, effective resource management is of utmost importance. The fusion of AI and EC has opened avenues for creative solutions to tackle the issues encountered in conventional agricultural sensor-cloud systems. This case study explores

the pioneering work presented by Rituparna Saha et al. [15]. The research introduces an innovative sensor-cloud architecture enhanced by EC and a sophisticated virtual sensing scheme named DLSense, aimed at transforming precision agriculture.

4.3.5.1 Challenges Addressed

- *Latency and Resource Consumption:* Traditional agricultural sensor-cloud systems suffer from high latency and substantial energy and bandwidth consumption due to the periodic transmission of raw sensor data to the cloud.
- *Unreliable Sensors:* The reliance on working sensor nodes poses limitations, especially in regions with damaged or inactive sensors.
- *Privacy Concerns:* Sharing sensitive farming data with third-party service providers raises privacy issues.

4.3.5.2 Key Benefits

- *Empowering Edge Devices:* The research introduces a modified sensor-cloud architecture that harnesses the capabilities of edge devices as an intermediary layer for sensor virtualization. This approach reduces the latency in service provisioning and decreases resource usage.
- *DLSense Virtual Sensing:* DLSense, an intelligent virtualization scheme, employs correlation theory and distributed learning within edge devices to forecast sensor data. This enables the exchange of trained models instead of raw data, ensuring data privacy while enhancing service accessibility.

4.3.5.3 Results and Implications

The performance of the DLSense scheme was assessed through thorough simulations and an experimental case study in precision agriculture. The findings underscored several notable advantages:

- *Latency and Cost Reduction:* DLSense achieves an 81% reduction in latency and a 66% reduction in service cost compared to state-of-the-art methods. This leads to quicker and more cost-effective data processing and analysis.
- *Increased Service Availability:* DLSense enhances service availability by 39%, ensuring that precision agriculture systems continue to operate effectively, even in regions with sensor failures.

This case study sheds light on the potential of Edge AI sensor virtualization in revolutionizing precision agriculture. The integration of edge devices, AI, and DLSense not only addresses the challenges of latency, resource consumption, and sensor reliability but also ensures data privacy. The promising results of reduced energy consumption, latency, and service costs, along with increased service availability, showcase the transformative impact of this approach.

4.3.6 Enhancing Crop Yield Prediction in Smart Farms with Edge AI

Crop yield prediction is a critical technique in agriculture, and the advent of smart farms, empowered by the IoT, has ushered in a wealth of real-time environmental data like temperature and humidity. However, challenges arise when handling sensitive or private data that cannot be transmitted over the internet. This case study explores innovative work presented by Junyong Park et al. [16]. The study introduces a scalable data analysis framework that leverages EC to pre-process and analyze private data, ultimately improving crop yield estimation.

4.3.6.1 Challenges Addressed

- *Private Data Handling:* Protecting sensitive farm data while still enabling effective analysis is a significant challenge, especially when internet connectivity is limited.
- *Data Separation:* Separating the data analysis process into local edge analysis and centralized server analysis while maintaining accuracy poses technical challenges.

4.3.6.2 Key Benefits

- *Scalable Data Analysis:* The proposed framework adopts a layered approach where edge devices pre-process and analyze local data, including growth data from environment sensors. This local analysis helps estimate current crop potential based on growth state.
- *Efficient Data Transmission:* By transmitting only early harvest rate information to the central server, the framework reduces the complexity of data transmission while enabling the server to trend-line predict future harvest yields.

4.3.6.3 Results and Implications

The study conducted real-world experiments in a tomato farm, demonstrating the effectiveness of the layered data analysis framework:

- *Accuracy Comparable to Server Analysis:* The results show that the error rate achieved by edge analysis is comparable to server-only analysis, even with significantly reduced feature sets.
- *Improved Complexity:* The proposed framework significantly reduces complexity while maintaining accurate results, making it a practical solution for smart farms with limited internet connectivity.

The layered data analysis method introduced in this case study offers a promising solution for enhancing crop yield prediction in smart farms. By leveraging EC for local data analysis and transmitting only essential information to the central server, sensitive data can remain protected from the internet while still contributing to accurate yield estimates.

4.3.7 Enhancing Data Sensing in Smart Agriculture through EC

A data sensing framework that leverages the synergy of EC and the IoT has been developed to address the challenges of acquiring valuable data throughout the crop lifecycle cost-effectively. This case study examines the innovative research conducted by Rihong Zhang and Xiaomin Li [17], with a specific focus on a novel data-sensing strategy aimed at enhancing the efficiency and cost-effectiveness of data collection during various crop growth stages.

4.3.7.1 Challenges Addressed

- *Limited Data Value:* Existing methods struggle to capture high-value data that correlates with specific crop growth stages.
- *Data Correlation:* Weak data correlation hinders the ability to derive meaningful insights from collected data.
- *High Data Collection Costs:* Traditional data collection approaches are resource-intensive and costly.

4.3.7.2 Key Benefits

- *Data Sensing Strategy:* The proposed strategy comprises four distinct phases:
 - Crop growth stage division using Gath-Geva (GG) fuzzy clustering.

- o Prediction of current crop growth stages employing a Tkagi-Sugeno (T-S) fuzzy neural network.
- o Optimization of environmental sensing parameters based on Deng's gray relational analysis.
- o An adaptive sensing method has been devised to accommodate sensing nodes with constraints on their effective sensing area.
- *Improved Data Value*: The strategy significantly enhances the value of collected sensing data by aligning it with specific crop growth stages.

4.3.7.3 Results and Implications

The study conducted extensive experiments to evaluate the performance of the proposed data sensing strategy:

- *High Accuracy*: The results demonstrate that the strategy can effectively divide and predict crop growth stages with high accuracy.
- *Efficiency Gains*: The strategy substantially reduces data sensing and collection times, resulting in decreased energy consumption and lower data collection costs.
- *Enhanced Data Value*: By improving the alignment of collected data with crop growth stages, the strategy elevates the overall value of sensing data.

Smart agriculture requires a robust data monitoring system capable of cost-effectively collecting valuable data throughout the entire crop life-cycle. This case study introduces a data collection framework that integrates sensors, EC, and IoT. It presents a data sensing strategy based on EC principles, consisting of four phases. This strategy aims to overcome challenges related to data value, correlation, and collection costs, ultimately yielding more efficient and valuable data.

4.3.8 Advancing Vineyard Management through Edge AI

Recent years have witnessed a notable shift in agriculture, with growing recognition of the potential benefits of Information and Communications Technology (ICT) tools and techniques. Official reports advocating for improved product quality, quantity, and economic conditions have underscored the need for the farming sector to embrace these innovations. Consequently, the deployment of sensing devices in agriculture has

surged, paving the way for the emergence of smart agriculture. This evolving concept encompasses various activities, most notably field monitoring, which provides decision-making support for actions like irrigation and fertilization.

In addition to conventional sensing devices utilizing internet protocols for data transfer, the realm of smart agriculture incorporates crop models. These models harness sensor data to deliver enhanced insights and recommendations for farmers, ultimately enhancing the quality and quantity of agricultural production.

This case study delves into the innovative work presented by Sergio Trilles and his team [18]. The study centers on the development of a cost-effective sensorized platform aligned with the IoT paradigm. The platform focuses on monitoring meteorological conditions, primarily to aid in the detection of mildew disease in vineyards. Leveraging EC and drawing inspiration from advances in Geographic Information Science (GIScience), this platform was deployed in a vineyard located in Vilafamés, Spain.

4.3.8.1 Challenges Addressed

- *Cost Reduction:* The platform aims to optimize the timing of downy mildew disease control treatments, thereby minimizing the costs associated with unnecessary chemical applications.
- *Environmental Impact:* By accurately timing treatment applications, the platform contributes to reducing the ecological footprint by limiting chemical usage and soil contamination.
- *Labor Efficiency:* The system enhances labor efficiency by eliminating the need for unnecessary vineyard visits and treatments.

4.3.8.2 Key Benefits

- *Cost Savings:* The platform identifies the optimal moments to apply downy mildew control treatments, resulting in significant reductions in chemical expenses.
- *Environmental Sustainability:* By curbing chemical usage, the platform promotes a healthier environment and more sustainable agricultural practices.
- *Labor Optimization:* Farm labor is used more efficiently, as treatments are only administered when needed, freeing up working hours for other essential tasks.

4.3.8.3 Results and Implications

The sensorized platform, referred to as SEnviro, has demonstrated its effectiveness in monitoring vineyards. By harnessing IoT and EC paradigms, an enhanced version of this platform processes data locally, enabling immediate information computation upon data collection. This approach not only reduces data transmission time and energy consumption but also enhances decision-making accuracy.

The primary focus of the platform's application is the timely detection of downy mildew treatment opportunities. Through real-time analysis of meteorological data, the system offers valuable insights for disease control. While this case study primarily tested the IoT platform's functionality and did not aim to validate disease models, the results were promising.

4.3.9 Transforming Animal Welfare Monitoring in Agriculture with Edge AI Computing

Recent years have witnessed a significant shift in agriculture, driven by the emergence of smart sensing and computing technologies. Smart agriculture systems, while still relatively nascent, hold immense promise for revolutionizing crop and livestock farming practices. This case study, inspired by the research conducted by Marcel Caria and his team [19], focuses on the development of an open and cost-effective smart farm computing system for monitoring animal welfare. The system leverages EC, fog computing, and cloud connectivity to create a comprehensive solution that addresses the complex and multifaceted aspects of animal well-being.

4.3.9.1 Challenges Addressed

- *Limited Accessibility*: Many existing smart agriculture systems are closed for experimentation, limiting opportunities for innovation and customization, particularly in the domain of animal welfare monitoring.
- *Comprehensive Monitoring*: Animal welfare encompasses a wide range of parameters and factors, making it challenging to develop a holistic monitoring solution that caters to the diverse needs of stakeholders.
- *Cost-Efficiency*: Keeping costs low while ensuring effective monitoring of animal welfare is a critical consideration for the agricultural sector.

4.3.9.2 Key Benefits

The proposed smart farm computing system offers several key advantages:

- *Open and Low-Cost:* By adopting open-source principles and utilizing cost-effective Raspberry Pi devices as edge nodes, the system provides accessible and customizable solutions for animal welfare monitoring.
- *Fog Computing Layer:* The conceptual creation of a fog computing layer enables real-time data processing at the edge, reducing latency and enhancing responsiveness in monitoring various parameters.
- *Cloud Connectivity:* Seamless integration with cloud computing systems allows for centralized data storage, analysis, and remote access, providing a comprehensive overview of animal welfare.
- *Multi-parameter Monitoring:* The system effectively monitors and collects data on multiple parameters relevant to animal welfare, offering valuable insights to stakeholders.

4.3.9.3 Results and Implications

The proposed smart farm computing system, which combines EC with cloud connectivity, demonstrates its effectiveness in monitoring diverse parameters related to animal welfare. By leveraging Raspberry Pi devices as edge devices, the system creates a fog computing layer that processes data locally, ensuring real-time responsiveness.

This innovative approach to animal welfare monitoring not only addresses the immediate needs of farmers and livestock caretakers but also offers possibilities for broader stakeholder engagement. Animal welfare, a multifaceted concept, can be measured, collected, evaluated, and shared, creating opportunities for improving the well-being of animals and fostering high-tech innovations in the agricultural sector.

This case study showcases the transformative potential of Edge AI computing in the context of smart agriculture, specifically in monitoring and enhancing animal welfare. By providing accessible, cost-effective, and open solutions, it opens up new avenues for improving agricultural practices, promoting animal well-being, and driving technological advancements in the sector.

4.3.10 Farming with CareBro: An Edge AI–Enabled IoT Farming Assistant

In the wake of the post-COVID-19 era, agriculture faces a pivotal transformation. The new imperative lies in enhancing productivity and ensuring the safety of agricultural produce, all the while reducing reliance on traditional labor-intensive practices. Technology emerges as the beacon of hope, and a contactless, reliable, and secure approach is the need of the hour. “CareBro” is a ground-breaking solution designed to redefine and revolutionize farming practices. This IoT-based smart agriculture system, powered by EC, takes the reins of farm management, making it autonomous and remote. It integrates seamlessly with smart farm sensors, orchestrating activities such as crop yield optimization, ethical pest management, and precision irrigation. CareBro operates as a cutting-edge solution, connected to the farmer via the cloud, offering real-time monitoring and intelligent decision-making. This innovation, authored by Atharv Tendolkar and Ramya [20], represents an innovative farm management solution, applicable across diverse farming landscapes, from urban and rural settings to large-scale and small-scale operations.

4.3.10.1 Challenges Addressed

- *Labor Shortage:* CareBro steps in as a solution for farming challenges exacerbated by labor shortages, ensuring the continuity of farm operations even during crises.
- *Productivity Enhancement:* The system aims to significantly increase crop yields while ensuring sustainable and ethical farming practices.
- *Precision Agriculture:* CareBro tackles the need for precision in pest management and irrigation, optimizing resource usage and minimizing environmental impact.
- *Remote Monitoring:* With real-time monitoring and remote control, it bridges the gap between farmers and their fields, offering a layer of convenience and control.

4.3.10.2 Key Benefits

- *Maximum Yield:* CareBro empowers farmers to maximize crop yields while providing valuable insights into challenges and mitigation strategies, all managed remotely and securely.

- *Adaptability*: The system is designed for adaptability, with sensors calibrated through beta testing to respond effectively to changing environmental conditions.
- *Multi-Functionality*: CareBro's potential extends beyond farm management; it can serve as a central hub for various farming services, including automated tasks like combine harvesting, nutrient spraying, and irrigation.
- *EC*: With its IoT network grid and local EC capabilities, CareBro enhances intelligence at every node across the entire farm.

4.3.10.3 Results and Implications

The development of CareBro has showcased its potential to revolutionize farming practices. Beta testing and sensor calibration have ensured the device's adaptability to dynamic environments. Integrating multiple modules and enhancing connectivity through a physical chassis resembling a scarecrow has been a crucial step. CareBro's scalability and multi-functionality, capable of serving as a central hub for various farm services, hold significant promise for the future. With its wide-ranging applicability, economic viability, and potential for scalability, CareBro emerges as a quintessential farm assistant for the modern, IoT-driven farming landscape.

CareBro operates sustainably, offering convenience and precision in farm management, fostering trust in technology. As technology continues to evolve, with the advent of 5G and advancements in EC, CareBro is poised to meet farmers' evolving needs and expectations. It stands as a testament to customer-centricity, sustainable farming practices, ethical principles, and safety.

4.4 DISCUSSION

The case studies presented in this chapter offer a rich tapestry of insights into the world of Edge AI applications in agriculture. These innovative systems bring to the forefront a multitude of functional features, utilization scenarios, benefits, socio-economic impacts, and adoption considerations that collectively define their profound significance to stakeholders in the agricultural domain.

4.4.1 Functional Features and Utilization

- *Real-Time Monitoring:* Edge AI systems excel in real-time monitoring of crucial agricultural parameters. This feature is indispensable in modern farming, as it allows farmers to obtain up-to-the-minute information on soil conditions, weather patterns, and crop health. The utilization of sensors, cameras, and data analytics enables the continuous collection and interpretation of data. As a result, farmers can make prompt decisions regarding irrigation, pest control, and disease management.
- *Data-Driven Decision-Making:* These systems leverage advanced analytics and machine-learning algorithms to generate data-driven recommendations. By analyzing large datasets collected from farms, they provide actionable insights for precise resource allocation. For example, they can determine the exact amount of water required for irrigation, the optimal timing for pesticide application, or the need for specific nutrients. This data-driven decision-making not only enhances crop yields but also minimizes the environmental impact of farming practices.
- *Interconnectivity:* Accessibility is a cornerstone of Edge AI systems. They are designed for seamless interconnectivity, allowing farmers to monitor and manage their farms remotely. Mobile applications and web interfaces provide user-friendly access to real-time data and control capabilities. Farmers can receive alerts and notifications, view reports, and adjust settings through these interfaces, regardless of their physical location. This functionality enhances convenience and efficiency in farm management.
- *Security and Privacy:* Data security and privacy are paramount. To address these concerns, some systems incorporate blockchain technology. Blockchain ensures the integrity and security of data by creating tamper-proof records. Furthermore, these systems often prioritize local data processing at the edge, meaning that sensitive information is processed on-site rather than being transmitted to centralized servers. This approach reduces the risk of data breaches during transmission.

4.4.2 Benefits

- *Resource Efficiency:* Edge AI systems are champions of resource efficiency. By continuously monitoring soil moisture, temperature, humidity, and other parameters, they optimize resource utilization.

This optimization extends to water, fertilizers, pesticides, and energy. For instance, the precise control of irrigation systems reduces water wastage, which is particularly critical in regions facing water scarcity. Similarly, tailored fertilization and pest management reduce chemical runoff, contributing to environmental sustainability. These efficiency gains result in cost savings for farmers.

- *Increased Crop Yield:* Timely interventions facilitated by Edge AI systems translate into increased crop yields. By detecting diseases, pests, or water stress at an early stage, these systems enable farmers to take swift corrective actions. For instance, they can trigger irrigation when soil moisture drops below a certain threshold, preventing crop wilting. Moreover, they can identify the presence of pests or diseases through image recognition, allowing for targeted treatments. The cumulative effect of these measures is higher crop productivity, ultimately benefiting both food security and farm profitability.
- *Data-driven Insights:* Edge AI systems provide farmers with valuable data-driven insights. As climate patterns become more unpredictable and farming practices evolve, these insights are crucial for making informed decisions. Farmers gain access to historical and real-time data, allowing them to adapt to changing conditions. For example, they can adjust planting schedules based on weather forecasts or optimize irrigation regimes according to soil moisture levels. This data-driven approach fosters resilience in agriculture, enabling farmers to navigate challenges effectively.

4.4.3 Socio-economic Impact

- *Empowering Smallholders:* In regions characterized by a high prevalence of smallholder farmers, Edge AI systems play a pivotal role in leveling the playing field. These technologies provide access to advanced farming practices and tools that were once beyond the reach of many small-scale farmers. As a result, rural communities benefit from enhanced agricultural productivity and improved livelihoods. The socio-economic impact is particularly significant in developing countries where agriculture is a primary source of income.
- *Environmental Sustainability:* The resource-efficient practices promoted by Edge AI systems contribute to environmental sustainability. Reduced water usage, minimized chemical runoff, and lower energy consumption collectively mitigate the environmental impact

of agriculture. These systems align with global sustainability goals by promoting eco-friendly farming practices. The socio-economic implications extend to cleaner environments, healthier ecosystems, and a reduced carbon footprint.

4.4.4 Adoption and Future Outlook

- *Scalability*: One of the key drivers of adoption is the scalability of Edge AI systems. They are designed to accommodate multiple users and farms simultaneously. This scalability makes them accessible to a broad spectrum of stakeholders, from individual farmers to large agricultural enterprises. As more farms adopt these technologies, the collective impact on agriculture becomes increasingly significant.
- *Interdisciplinary Integration*: Edge AI systems demonstrate versatility through interdisciplinary integration. They incorporate various technologies, including IoT, blockchain, and cameras for pest and plant health monitoring. This adaptability positions them for further growth and adaptation to evolving agricultural challenges. For instance, the integration of cameras for pest detection and plant health monitoring expands their utility beyond resource management to include proactive pest control.

4.5 CONCLUSION

In this chapter, a journey through the dynamic landscape of Edge AI in agriculture was embarked upon. The fundamental concept of Edge AI was explored, its potential was deciphered, and its pivotal role in reshaping modern agriculture was recognized. A new era of immediacy, responsiveness, and efficiency in farming practices is heralded by the integration of AI with EC and the IoT.

As a series of case studies was delved into, how Edge AI has been applied to address the most pressing challenges in agriculture was witnessed, spanning both national and international contexts. The transformative potential of Edge AI was demonstrated by these real-world examples, showcasing its versatility and adaptability. From precision agriculture and resource optimization to livestock management and crop monitoring, Edge AI was proven to be a catalyst for innovation, enabling farmers

and stakeholders to make data-driven decisions, reduce waste, and enhance productivity.

The adoption of Edge AI in agriculture is not without its challenges, including issues related to connectivity, data privacy, and scalability. However, solutions are met that ensure seamless integration and the safeguarding of sensitive agricultural data.

In summary, a powerful ally in the pursuit of sustainable and efficient farming practices is Edge AI. Stakeholders are empowered with real-time insights, enabling informed decisions that benefit both the agricultural industry and the environment. As we move forward, the potential for innovation in Edge AI remains boundless, promising a future where agriculture becomes smarter, greener, and more resilient.

REFERENCES

- [1] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision and challenges," *IEEE Internet Things J.*, vol. 3, no. 5, pp. 637–646, 2016.
- [2] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing: The communication perspective," *IEEE Commun. Surv. Tutor.*, vol. 19, no. 4, pp. 2322–2358, 2017.
- [3] D. Situnayake and J. Plunkett, *AI at the edge: Solving real-world problems with embedded machine learning*. Sebastopol, CA: O'Reilly Media, 2023.
- [4] Z. Zhou, X. Chen, E. Li, L. Zeng, K. Luo, and J. Zhang, "Edge intelligence: Paving the last mile of artificial intelligence with edge computing," *Proc. IEEE Inst. Electr. Electron. Eng.*, vol. 107, no. 8, pp. 1738–1762, 2019.
- [5] T. H. Luan, L. Gao, Z. Li, Y. Xiang, G. Wei, and L. Sun, "Fog computing: Focusing on mobile users at the edge," *arXiv [cs.NI]*, 2015.
- [6] J. Wang, Z. Feng, S. George, R. Iyengar, P. Pillai, and M. Satyanarayanan, "Towards scalable edge-native applications," in *Proceedings of the 4th ACM/IEEE Symposium on Edge Computing*, 2019.
- [7] F. J. Ferrández-Pastor, J. M. García-Chamizo, M. Nieto-Hidalgo, J. Mora-Pascual, and J. Mora-Martínez, "Developing Ubiquitous Sensor Network platform using Internet of Things: Application in precision agriculture," *Sensors (Basel)*, vol. 16, no. 7, p. 1141, 2016.
- [8] S. T. Oliver, A. González-Pérez, and J. H. Guijarro, "An IoT proposal for monitoring vineyards called SEnviro for agriculture," in *Proceedings of the 8th International Conference on the Internet of Things*, 2018.
- [9] D. Berckmans, "General introduction to precision livestock farming," *Anim. Front.*, vol. 7, no. 1, pp. 6–11, 2017.
- [10] X. Zhang, Z. Cao, and W. Dong, "Overview of edge computing in the agricultural internet of things: Key technologies, applications, challenges," *IEEE Access*, vol. 8, pp. 141748–141761, 2020.

- [11] H. T. Minh, T. P. Anh, and V. N. Nhan, "A novel light-weight DCNN model for classifying plant diseases on internet of things edge devices," *Mendel*, vol. 28, no. 2, pp. 41–48, 2022.
- [12] L. Ting, M. Khan, A. Sharma, and M. D. Ansari, "A secure framework for IoT-based smart climate agriculture system: Toward blockchain and edge computing," *J. Intell. Syst.*, vol. 31, no. 1, pp. 221–236, 2022.
- [13] X. Li, Z. Ma, J. Zheng, Y. Liu, L. Zhu, and N. Zhou, "An effective edge-assisted data collection approach for critical events in the SDWSN-based agricultural internet of things," *Electronics (Basel)*, vol. 9, no. 6, p. 907, 2020.
- [14] D. Shadrin, A. Menshchikov, D. Ermilov, and A. Somov, "Designing future precision agriculture: Detection of seeds germination using artificial intelligence on a low-power embedded system," *IEEE Sens. J.*, vol. 19, no. 23, pp. 11573–11582, 2019.
- [15] R. Saha, A. Chakraborty, S. Misra, S. K. Das, and C. Chatterjee, "DLSense: Distributed learning-based smart virtual sensing for precision agriculture," *IEEE Sens. J.*, vol. 21, no. 16, pp. 17556–17563, 2021.
- [16] J. Park, J.-H. Choi, Y.-J. Lee, and O. Min, "A layered features analysis in smart farm environments," in *Proceedings of the International Conference on Big Data and Internet of Thing*, 2017.
- [17] R. Zhang and X. Li, "Edge computing driven data sensing strategy in the entire crop lifecycle for smart agriculture," *Sensors (Basel)*, vol. 21, no. 22, p. 7502, 2021.
- [18] S. Trilles, J. Torres-Sospedra, Ó. Belmonte, F. J. Zarazaga-Soria, A. González-Pérez, and J. Huerta, "Development of an open sensorized platform in a smart agriculture context: A vineyard support system for monitoring mildew disease," *Sustain. Comput. Inform. Syst.*, vol. 28, no. 100309, p. 100309, 2020.
- [19] M. Caria, J. Schudrowitz, A. Jukan, and N. Kemper, "Smart farm computing systems for animal welfare monitoring," in *2017 40th International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO)*, 2017.
- [20] A. Tendolkar and S. Ramya, "CareBro (personal farm assistant): An IoT based smart agriculture with edge computing," in *2020 Third International Conference on Multimedia Processing, Communication & Information Technology (MPCIT)*, 2020.